EMI 2026 · ENGINEERING MECHANICS INSTITUTE

HyperNetwork with Transformer-Based Weight Generation for Regional Wind-Induced Structural Response Estimation

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THE PROBLEM

Portfolio-scale wind resilience is computationally prohibitive.

0 s
OpenSees per building single CPU core

0
buildings in a regional portfolio

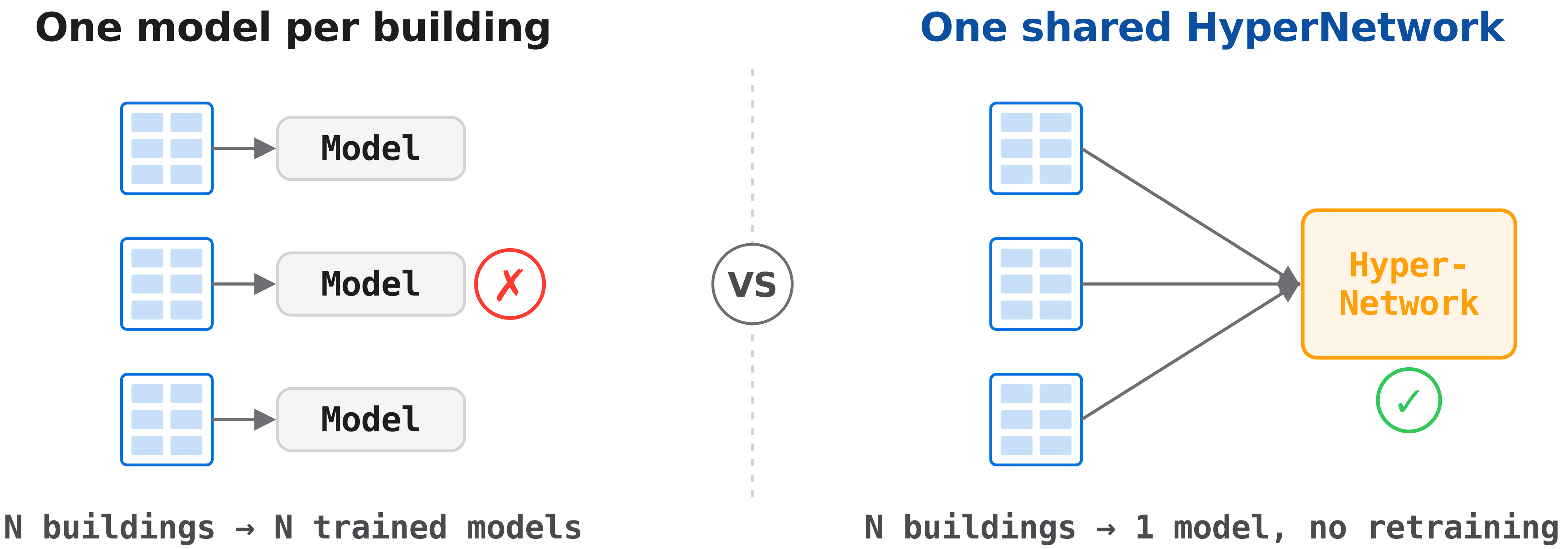
0 h
wall-clock to simulate 900 test buildings

One nonlinear time-history analysis per building, per hazard scenario. Multiply by thousands of structures — direct simulation is impractical at portfolio scale.

Portfolio-scale wind assessment spans large, heterogeneous building inventories

THE GAP

Conventional surrogates don't transfer across buildings.

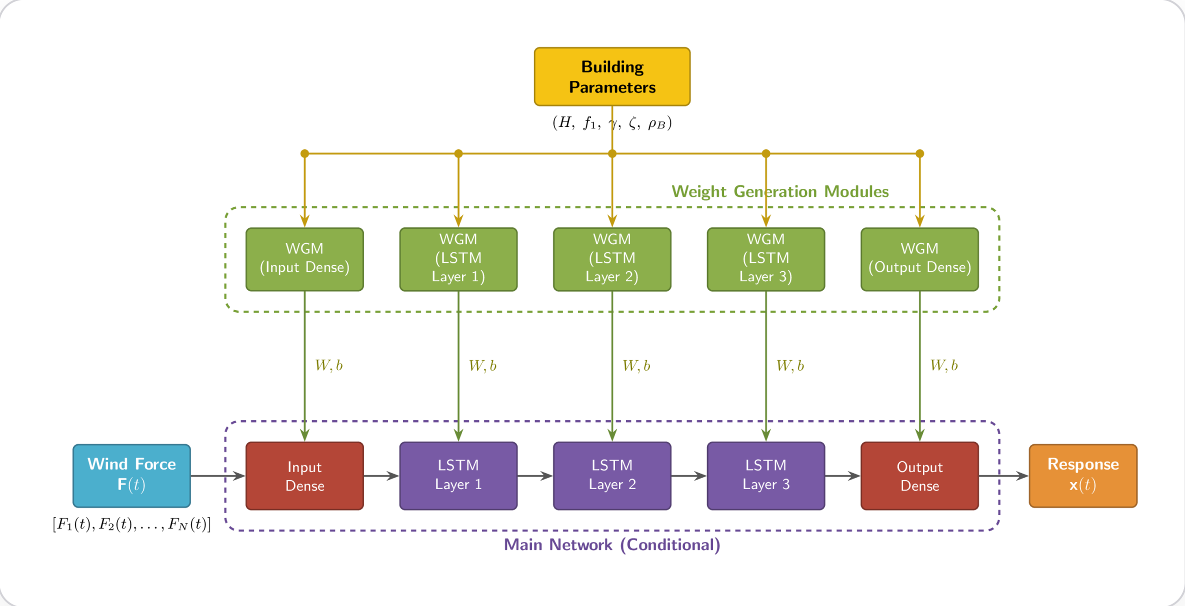


Different **height, stiffness, damping, mass** —

one surrogate per building doesn't transfer

KEY IDEA

Generate the response network's weights from the building's feature vector.



One HyperNetwork reads building features and generates a dedicated response predictor — **one trained model, any building**.

ARCHITECTURE

From features to response — one forward pass.

Two inputs: building features generate the network; the wind-load sequence drives it to produce the response.

Transformer WGM

2 layers · dim 64 · 2 heads · FFN 256

Dense → LSTM×3 → Dense

Weights generated, not stored

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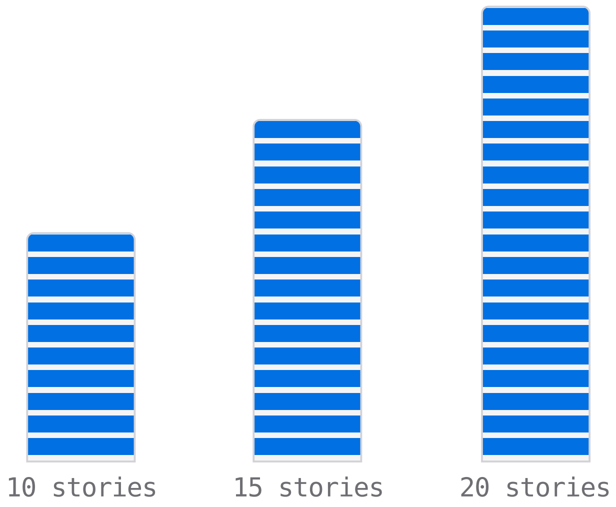
HEIGHT NORMALIZATION

10 / 15 / 20 stories — one shared 20-point grid.

Map each building's floor responses to a fixed normalized-height axis so every output is always active.



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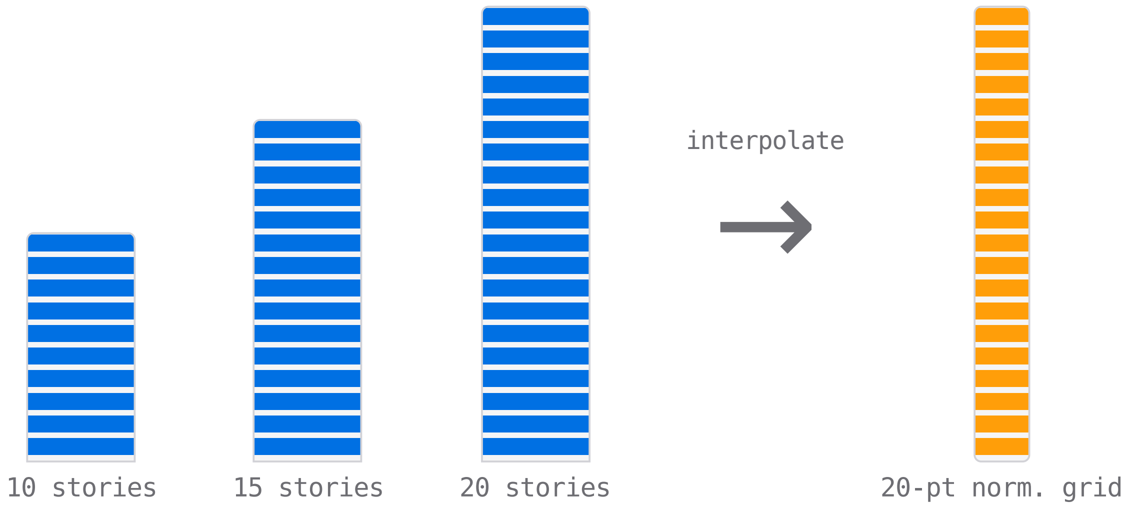

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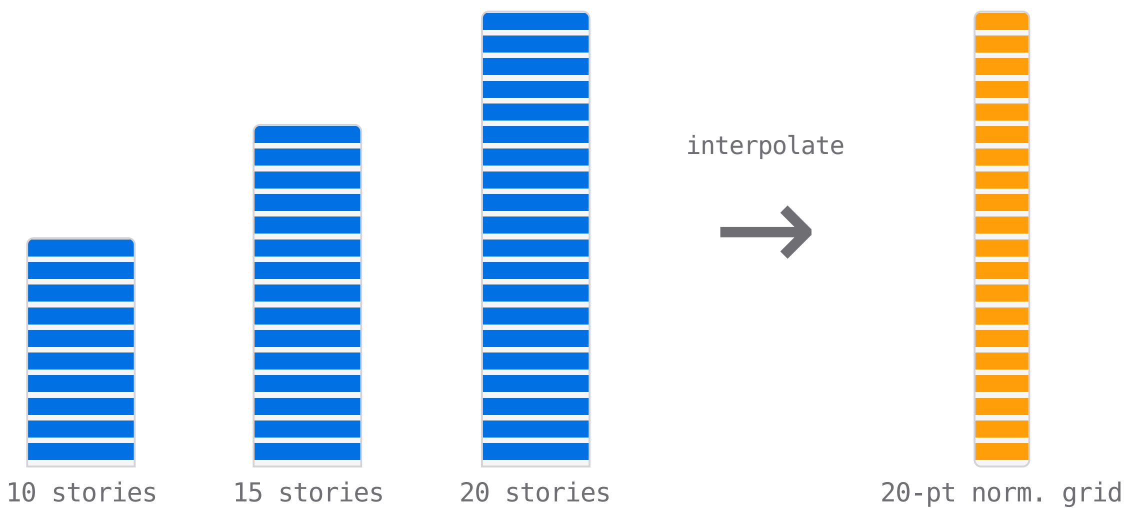

Zero-padding: inactive floors waste capacity

Interpolation: all 20 outputs always active

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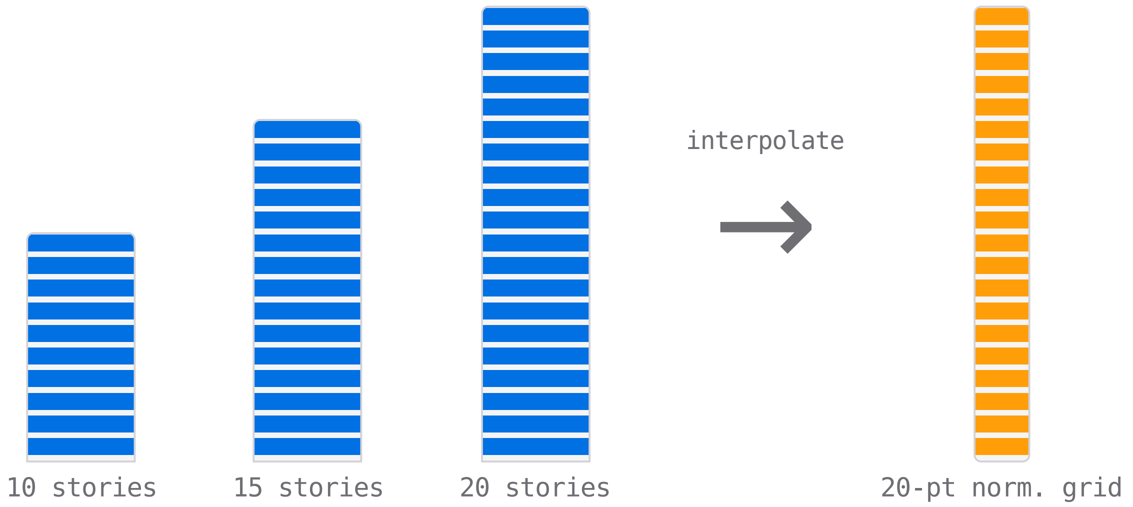


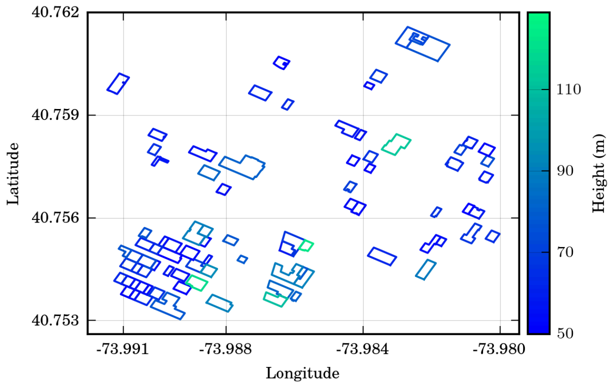
Zero-padding: inactive floors waste capacity

Interpolation: all 20 outputs always active

6.83 M parameters · population mean NRMSE: 1.05 % (10-story) · 1.01 % (15-story) · 0.88 % (20-story) — error is uniform across story counts

EXPERIMENTAL SETUP

Times Square–calibrated benchmark — scaled to 9,000.



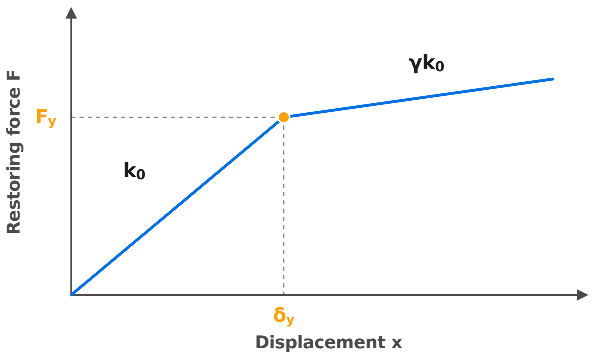
85 real buildings near Times Square (**OpenStreetMap**, 50–123 m) — the reference for the synthetic benchmark.

160 synthetic cases · 128/16/16 split

80 configs × 2 wind realizations

60 s wind records

BILINEAR STIFFNESS



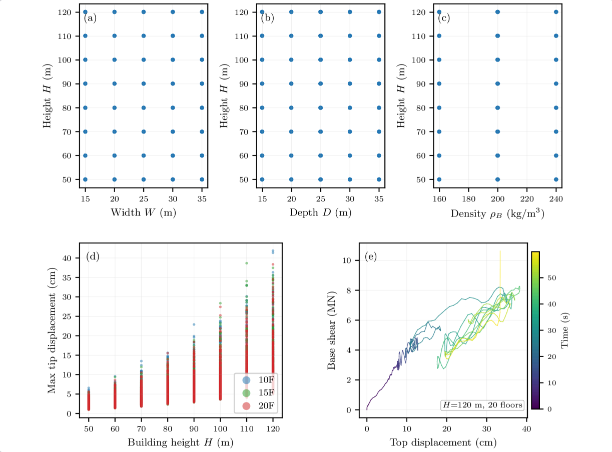
F restoring force · **x** displacement

k_0 initial (elastic) stiffness

F_y yield force · **δ_y** yield displacement

γ post-yield ratio $\in [0.05, 0.10]$ · **γk_0** post-yield stiffness

9,000-BUILDING FEATURE GRID



H height · **W** width · **D** depth

ζ damping ratio · **ρ** density

N_s number of stories · **f** fundamental frequency

Full-factorial over 7 features; training (blue) spans the test portfolio (red).

160-CASE TIMES SQUARE BENCHMARK

NRMSE 0.017 on 16 held-out test cases.

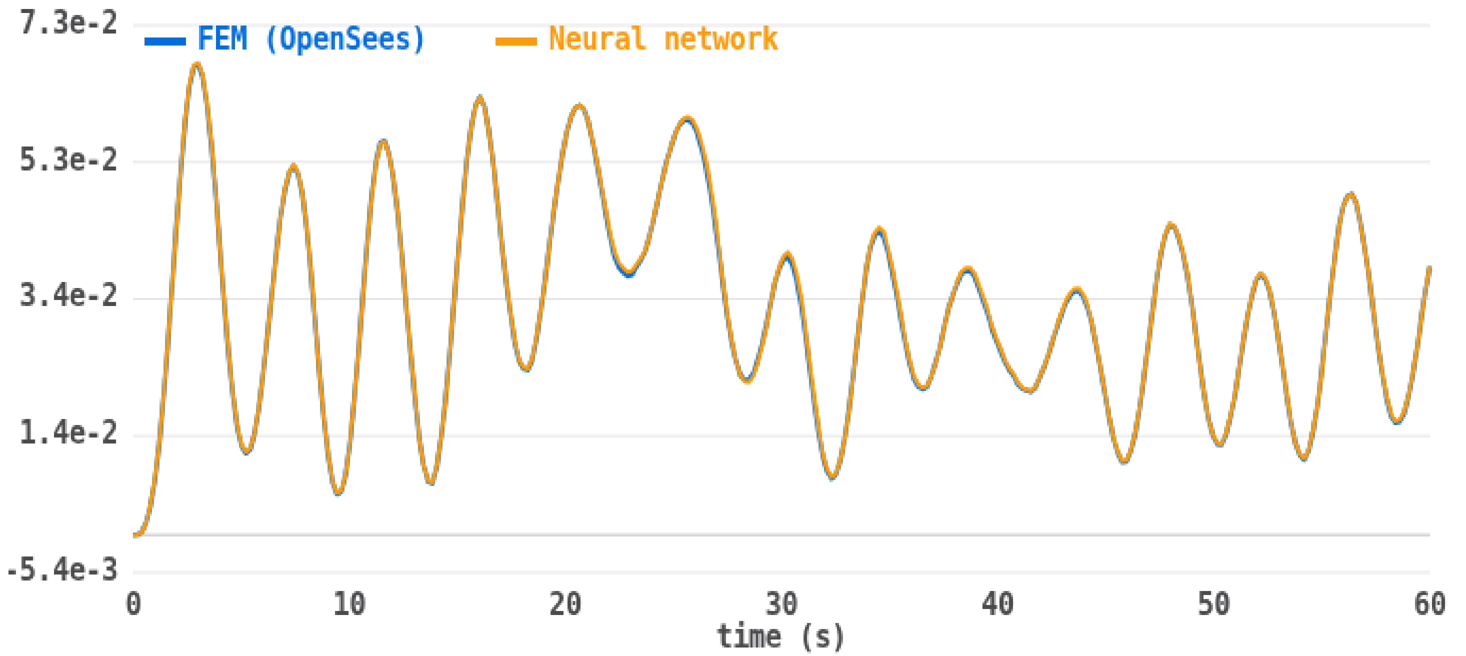
MEAN NRMSE

0.000

16 held-out test cases

160 synthetic cases — 128 train / 16 val / **16 test**
80 configs × 2 wind realizations. No test-case simulation in training.

LOAD CASE **Wind** Constant Sine Impulse Play



FEM HyperNetwork Demonstration building — H = 85 m, 15 floors, top-floor displacement

SCALE & GENERALIZATION — 9,000-BUILDING DATASET

Under 1% NRMSE on 900 held-out buildings.

0.00%
mean NRMSE
900 held-out buildings

0.00%
median NRMSE
900 held-out buildings

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training buildings
building-level split

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UNIFORM ACROSS HEIGHTS

1.05%
10 floors

1.01%
15 floors

0.88%
20 floors

Height-normalized interpolation keeps error even across all building heights.

DURATION GENERALIZATION

60 s
training window

→

600 s
evaluated signal

Separate long-duration test — error stays low throughout the full 600-s signal. Model learned structural dynamics, not a fixed template.

~7,000× faster than single-core OpenSees.

OpenSees FEM	~80 s / building
Neural network	~11.4 ms / building (batched)

~0x

faster per building

0 s OpenSees / building, single CPU core → **0×** HyperNetwork, batched GPU inference

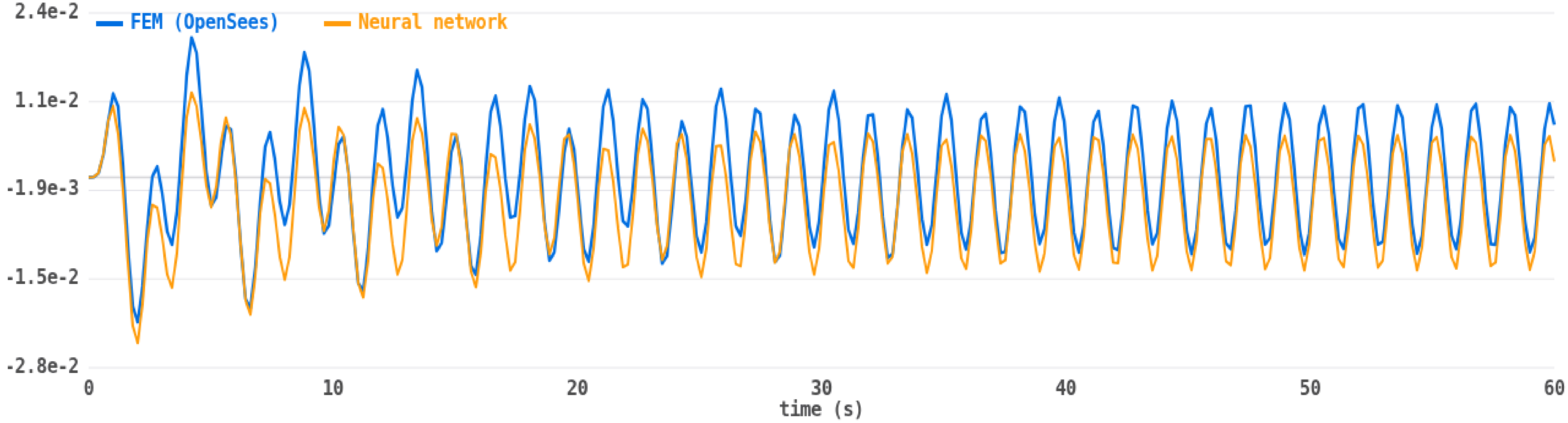
≈ 20 h OpenSees, portfolio of 900 → **seconds** HyperNetwork, same 900 buildings

LIVE VERIFICATION

Probe any load — surrogate vs. FEM, on demand.

Probe any load class live — the surrogate verified against a fresh FEM run on demand. No cached data. Every verdict is a new execution.

LOAD CASE Wind Constant Sine Impulse Play



Demonstration building (top-floor displacement) — H 85 m · 15F · ζ 0.02. Resonant sine shown. Switch load class with the controls above.

MULTI-INPUT, MULTI-OUTPUT

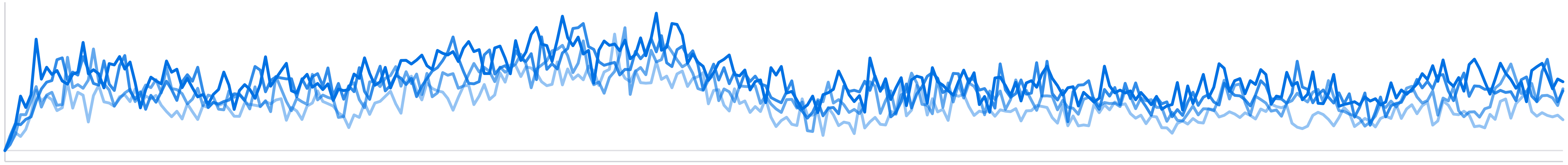
Multi-force input → multi-response output.

The surrogate is MIMO: forces at every floor map to displacements at every floor, matched against FEM.

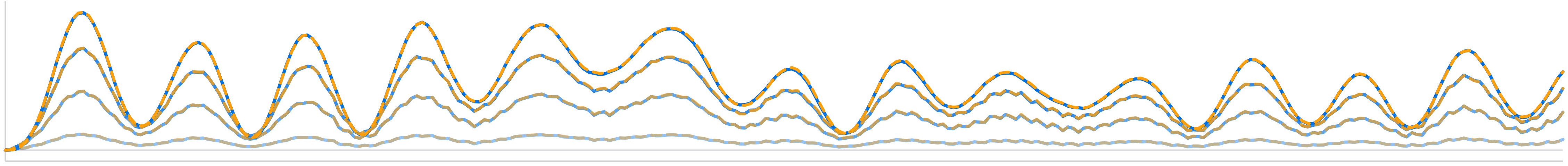
LOAD CASE Wind Constant Sine Impulse Play

Input — applied forces

per-floor — non-uniform



Output — floor displacements



BEYOND WIND

Complex loads — strong results, one honest limit.

Population mean NRMSE — 900 held-out buildings:

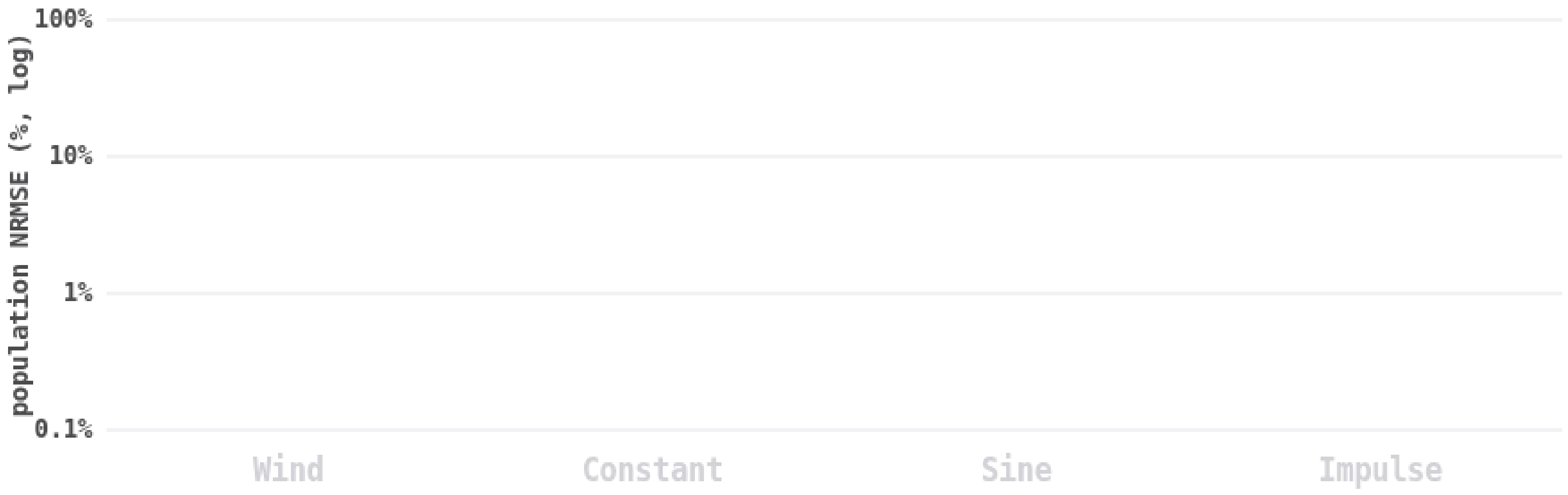
1.25 %
Wind

4.6 %
Constant

13.7 %
Sine (near-resonance)

54.2 %
Impulse — current limit

Impulse shown honestly. More complex loads are active future work.



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LOOKING AHEAD

Future directions.

- **More complex loads — the immediate next step.** Improve impulse coverage; extend to blast, multi-hazard, and arbitrary broadband excitations.
- **Physics-informed / knowledge-enhanced training** — embed governing equations to improve extrapolation and cut data demand for larger, more diverse inventories.
- **Richer features + 3D models** — image and BIM-driven building representations; extend to 3D FEM with torsional and complex nonlinear behavior.

TAKEAWAYS · IMPACT

One model, many buildings.

- **Cross-building generalization** — one trained model serves a heterogeneous portfolio with no per-building retraining.
- **~7,000× faster** than OpenSees (~80 s/building, single CPU core).
- **<1 % median NRMSE** on 900 held-out buildings (0.89 %).
- **Portfolio resilience + digital twins** — regional screening and scenario analysis now tractable on ordinary GPU hardware.

7,000×

speedup over OpenSees

0.89 %

median NRMSE · 900 test buildings

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Thank you.

Questions?

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